Enlace

NETWORK TELEMETRY PILOT · PROPOSAL

See your fibre network before your customers do.

A read-only telemetry agent that reads the data your OLTs already produce and turns it into early warnings — fibre cuts, degrading signal, optical faults, ghost connections and customers drifting toward churn.

Prepared for **Community Fibre**

By **Pulso Technologies Limited** — a UK telecom-technology startup
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ABOUT THIS PROPOSAL

Pulso Technologies is a **pre-launch telecom-technology startup**, and Enlace is our **first product**. This document sets out what Enlace does and what we **intend to deliver** in a pilot with you. Every result shown here comes from our own **internal validation** on representative data — not customer data. The pilot is how we prove it on your live network.

THE OPPORTUNITY

Your network is already talking. Most NOCs can't hear it.

For a fast-scaling London full-fibre network like Community Fibre, visibility has to keep pace with every premises passed. Your OLTs already know when a customer's signal is dying, when a PON branch has gone dark, and when a connection has quietly stopped carrying traffic — but that data sits siloed by vendor, unread, so operations stay **reactive**.

The customer calls first. A ticket opens. An engineer rolls a van — often to find no fault at the premises at all. Reactive operations cost twice: in truck rolls (£80–150 each) and in the churn of customers who lost service before anyone knew. **Enlace closes that gap.** One lightweight agent reads telemetry from the equipment you already run, normalises it across vendors, and runs the analysis below at the edge — flagging the specific problem, on the specific ONT, before the phone rings.

AT A GLANCE

Eleven things Enlace finds in your network.

Fibre cuts
Mass dropouts on a PON branch, with an estimated break location

Power vs fibre
Dying-gasp signatures separate an area power cut from a physical break

Reflectance culprit
The one ONT whose return-loss is knocking its neighbours offline

Signal degradation
Per-ONT optical decline with an estimated time-to-failure

Optical budget
Loss attributed to fibre, splitter, connectors — with the probable cause

SFP ageing
OLT transceiver wear, seen as a whole-port synchronised decline

Weather faults
Condensation, rain ingress and thermal stress — no weather station needed

Flapping ONTs
Repeated up/down cycling, with cascade-risk and probable cause

Ghost connections
Provisioned, online, but carrying no real traffic — revenue leakage

Churn risk
90-day churn probability and revenue at risk, from signal behaviour

Capacity
Splitter ports trending toward exhaustion, with months-to-full

Help-desk view
Colour-coded health your front line can read without an engineer

Each one is normally invisible until it becomes a support call or a repeat truck roll. Enlace surfaces them as prioritised **tickets** with a fix, a team and an ROI — and an executive **health score** for the whole network.

HOW IT WORKS

One agent: collect, detect at the edge, act.

01 · COLLECT

From the kit you already run

Read-only polling of OLTs, MikroTik routers, RADIUS and CPE — normalised into one data model across every vendor.

02 · DETECT

Analysis on-site

Eleven detection modules run locally on the agent. No raw customer data leaves the network — only findings.

03 · ACT

Tickets & dashboards

Prioritised tickets to your NOC, alerts to Slack/Elasticsearch, and a help-desk health view — or just a report.

Speaks every box on your network

A single binary (~6 MB, <1% CPU) collects over seven protocols and normalises twelve OLT vendors plus a generic SNMP fallback. Polling is configurable, as fast as 60-second intervals. If the link to the cloud drops, readings buffer locally and sync later.

PROTOCOLS

`SNMP v2c/v3` `SSH CLI` `NETCONF` `gRPC streaming` `RouterOS API` `RADIUS` `TR-069 / GenieACS`

OLT VENDORS

`Adtran SDX` `Huawei` `Nokia` `ZTE` `FiberHome` `Datacom` `Intelbras` `Parks` `BDCOM` `VSOL` `CDATA` `Ubiquiti` `+ generic SNMP`

Per ONT, Enlace normalises: serial, PON port, Rx/Tx optical power, status, distance, uptime, last-down cause and Ethernet link speed — plus temperature, voltage and laser bias where the equipment exposes OMCI/NETCONF detail.

FAULT INTELLIGENCE

Know what broke, why, and roughly where.

Mass-offline detection — fibre cut vs power outage

When several ONTs on a PON port drop together, Enlace classifies the root cause by whether they sent a dying gasp first. No dying gasp → a physical fibre cut. Dying gasps → an area power failure. A mix → both, flagged as such.

`Trigger: ≥ configurable N offline within a 60s window · severity scales with affected count`

Reflectance — the "return-loss" culprit ONT

A dirty connector or damaged fibre on one ONT can throw optical reflections that corrupt upstream timing and knock its neighbours offline — while the culprit itself stays up. Enlace finds the offending ONT: it isolates the mass-dropout, rules out fibre cuts and already-degraded units, then looks for the Tx-power anomaly on the ONT that remained online.

`Output: suspect serial, distance, Tx anomaly (dBm) · confidence 0.3 → 0.9 as the same suspect recurs across events`

Fault localisation — narrow the search before the van rolls

Using ONT distances (and PON-tree topology where available), Enlace estimates where on the fibre run the break sits — e.g. "break at 800–850 m on port 0/1/4" — so the field team starts at the right joint instead of walking the whole route.

`Output: estimated distance from OLT + human-readable span (last-online → first-offline node)`

PREDICTIVE & OPTICAL HEALTH

Catch failures while they're still trends.

Signal degradation & time-to-failure

Enlace fits a trend to each ONT's optical power over a rolling 30-day history and projects when it will cross the failure threshold — turning "it broke last night" into "this one fails in ~9 days, schedule it." Every forecast carries a confidence (the fit's R^2).

`Severity by slope (dBm/day): Watch -0.015 · Warning -0.035 · Critical -0.07 or Rx < -27 dBm · output: days-to-failure + confidence`

Optical budget — attribute the loss

For each ONT, Enlace models the expected loss (fibre attenuation, splitter, connectors) and compares it to reality, attributing any excess loss to a probable cause — so a marginal link comes with a hypothesis, not just a number.

Margin to threshold	Status	Excess loss points to…
> 6 dB	Excellent	—
3–6 dB	Good	—
1–3 dB	Marginal	Connector loss · splice degradation · macro-bend
0–1 dB	Critical	Splitter over-loss · insufficient OLT power
< 0 dB	Failed	Below receiver sensitivity

SFP ageing on the OLT

When an OLT transceiver wears out, every ONT on that port drifts down together, regardless of distance. Enlace spots that synchronised, port-wide decline and flags the port as an outlier against its siblings — distinguishing a £40 SFP swap from a fibre job.

`Output: per-port Rx trend (dBm/week), how tightly ONTs track it, weeks-to-failure, outlier-vs-siblings flag`

ENVIRONMENTAL & STABILITY

Faults the weather and loose joints cause.

Weather-correlated faults — no weather station required

By grouping ONTs that sit close together on a span, Enlace separates infrastructure problems from one-off ONT faults and reads the environmental signature directly from the optics:

- **Nighttime condensation** — signal consistently worse overnight → moisture in a splice enclosure or aerial joint.
- **Periodic rain ingress** — coordinated 2–8 hour dips that fully recover → a failed water seal in a joint.
- **Thermal expansion** — higher signal variance in afternoon heat → aerial cable under thermal stress.

`Requires ≥ 3 ONTs within 100 m on a port · output: pattern, affected ONTs, distance range, correlation strength`

Flapping ONTs & cascade risk

An ONT cycling online/offline wastes OLT CPU, drowns the NOC in alarms, and on a busy port can cascade-crash the whole branch. Enlace measures the flap rate, scores the cascade risk by how loaded the port is, and proposes a probable cause.

`Severity: Sporadic 2-5/h · Moderate 5-15/h · Severe >15/h · cause: marginal Rx (dirty connector), power instability, or periodic hardware fault`

CUSTOMER & REVENUE INTELLIGENCE

Connect the network to the P&L.

Ghost connections — find the revenue leakage

Some ONTs are provisioned and "online" but never actually used. Enlace flags them — either no Ethernet link at all, or a healthy signal with a perfectly flat profile (real customers create variance) — and estimates the revenue tied up in each.

`Criteria: uptime > 90%, healthy Rx, no Ethernet activity / flat variance · sorted by days online (biggest leak first)`

Churn risk & revenue at risk

Enlace translates optical behaviour into a 90-day churn probability — combining the signal trend with "micro-dropouts" (brief disconnections that frustrate customers) — and attaches the monthly revenue at risk, so retention can act on the right accounts first.

`Impact tiers from Rx + slope (and a bump for repeated micro-dropouts) · output: probability, days degrading, revenue at risk`

Capacity planning

Enlace infers each splitter's size, measures how fast it's filling, and projects when it hits 95% — so you order and build ahead of demand instead of scrambling when a street sells out.

`Alerts: Watch 50-75% · Warning 75-90% · Critical >90% · output: splitter type, utilisation, new connections/month, months-to-full`

FROM INSIGHT TO ACTION

Findings become tickets, not noise.

Every detection is turned into a structured ticket — with a priority, the right team, an SLA, an estimated fix cost and the revenue at risk, so the NOC can triage by ROI rather than by whoever shouted loudest.

Finding	Priority	Routed to	SLA
Reflectance source ONT	P1	Customer / field	1 day
Capacity critical	P2	Passive (build)	5 days
Churn risk cohort	P2	Management / retention	5 days
Weather / water ingress	P3	Passive (splice)	14 days
Ghost connections	P3	Management	14 days

Impact scoring

Faults are weighted by how many customers are affected and when — a daytime business-hours outage outranks the same fault at 3 a.m. — so the highest-impact work floats to the top automatically.

Help-desk health view

A colour-coded card (green / amber / red) per OLT, written in plain language. A front-line agent can tell a caller "there's an area power fault affecting your street, crews are aware" — without escalating to an engineer or reading an SNMP table.

INTERNAL VALIDATION

What a single 7-day export reveals.

Internal validation run — representative data, not customer data: ~52 ONTs across 4 PON ports, one OLT, 7 days of optical/status readings. Enlace surfaced:

52
HEALTH SCORE

`1 fibre cut` — a PON branch where multiple ONTs dropped together with no dying gasp (a physical break, not power)

`1 area power event` — separated from the fibre cut by dying-gasp signatures

`1 ghost connection` — online all week, carrying no real traffic

`3 at-risk customers` — signal trending toward failure, scored for churn

`1 flapping ONT` — repeated instability on a marginal line

Engine: validated against a benchmark of real-world incident patterns (fibre cuts, splice and SFP pre-failure, reflectance, flapping, weather-correlated degradation, capacity exhaustion, churn) — all detected end-to-end; 180+ automated tests; audits ~1,000 ONTs in roughly 10 milliseconds. The pilot is how we validate it against your real network.

TRUST & INTEGRATION

Safe to run on a live network.

Read-only. Enlace never sends a write command to your equipment.

Credentials stay local. Only aggregated findings leave the network — never customer data.

No lock-in. Vendor-agnostic and built on open standards (SNMP, NETCONF) — your data stays yours.

Resilient. Local buffer survives connectivity loss; retries with backoff.

> ### Read-only by design — your engineers can verify it
> Enlace issues only query operations — SNMP `GET`, CLI `show`/`display` commands, NETCONF `<get>`, RouterOS read calls and a passive RADIUS listener. It never issues a set, write, configuration or reboot command. Equipment credentials stay on your network and are never transmitted — only computed metrics and findings leave. We'll walk your engineers through exactly what the agent does and every operation it runs, so they can sign it off — without us exposing our source.

Findings flow where your team already works: structured tickets, **Elasticsearch** for search and analytics, **Slack / PagerDuty** for P1/P2 alerts, or any custom webhook — and a dry-run mode that just prints, so you can see exactly what would be sent.

HOW THE PILOT WORKS

One CSV to first findings.

1. **Send us a month of data**
   Export roughly a month of ONT / optical-power readings from an area where you've had issues. No install, no agent on your network — just a CSV.

2. **We analyse it with you**
   We run it through Enlace and walk you through what it found — the faults, at-risk customers, ghosts and capacity hot-spots in that window, on your own data.

3. **Go live — read-only**
   When you're ready, we deploy the agent against a live OLT for the pilot — strictly read-only — so you see it work in real time. Free for the duration.

THE OFFER

A free pilot for launch partners.

We're inviting a small number of UK fibre operators to validate Enlace on their own networks — and we'd like **Community Fibre** to be one of them. The pilot is **free for the pilot period**, with preferential pricing reserved for launch partners when we move to paid.

The ask: one CSV export, or a 15-minute call. That's it.

Next step

Two ways to start, whichever is easier: book a **15-minute call** this week, or send us a single OLT export — about a month of data from an area you've had trouble with — and we'll return a findings report within **24 hours**. Email hello@enlace.network to set it up.

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